Optimization of Cement Distribution from Depots Using a Transportation Model

Submitted by: The TBA Group

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Case Study Information

The assigned case study is *Transportation Models for Cost Minimization in Distribution Logistics*, published in the *International Journal of Science, Strategic Management and Technology*, Volume 2, Issue 9 (September 2026). The article presents the distribution problem of MAISANGO Cement Nigeria Plc and applies transportation-model procedures to minimize distribution cost. The case uses four depots and five destinations and compares the Northwest Corner Method (NWC), Vogel's Approximation Method (VAM), Least Cost Method (LCM), and Row Minima Method (RMM) as procedures for obtaining initial feasible solutions. The article reports a final transportation cost of NGN 145,140. The underlying MAISANGO case is attributed in the assigned article to Abdullahi et al. (2020).

The Operations Research problem is a transportation model under linear programming. Its purpose is to determine how many cement bags should be shipped from each depot to each destination so that all available supply is allocated, all destination demand is satisfied, and total transportation cost is minimized. The model is appropriate for the case because each route has a specified transportation cost and the total supply is equal to total demand.

Case Study Summary

This case study focuses on the distribution of cement from four MAISANGO Cement Nigeria Plc depots, identified as Depots A, B, C, and D, to five destinations: Ikeja, Ajagunle, Apapa, Idi-Araba, and Agege. The main decision is the quantity of cement bags assigned to each possible depot–destination route. The objective is to minimize total transportation cost while satisfying all supply and demand requirements.

The total available supply is 5,100 cement bags, consisting of 1,100 bags from Depot A, 1,220 from Depot B, 950 from Depot C, and 1,830 from Depot D. The total demand is also 5,100 bags, consisting of 1,010 bags for Ikeja, 970 for Ajagunle, 980 for Apapa, 1,250 for Idi-Araba, and 890 for Agege. Since total supply equals total demand, the transportation problem is balanced.

Table 1: Transportation Cost, Supply, and Demand Data

Supply Center	**Ikeja**	**Ajagunle**	**Apapa**	**Idi-Araba**	**Agege**	**Total Supply**
A	68	36	26	31	62	1,100
B	35	72	36	28	54	1,220
C	40	55	41	30	28	950
D	25	35	40	80	34	1,830
Demand	1,010	970	980	1,250	890	5,100

Operations Research Model Replication

Below is the complete transportation model for distributing cement from the MAISANGO depots to the five destinations.

Operations Research Model

This is a transportation model of linear programming. It determines the number of cement bags to be shipped from each depot to each destination while minimizing total transportation cost. Because the problem is balanced, the model requires the entire available supply to be distributed and every destination's demand to be met exactly.

Sets and Decision Variables

Supply centers:

- Depot A
- Depot B
- Depot C
- Depot D

Destinations:

- Ikeja
- Ajagunle
- Apapa
- Idi-Araba
- Agege

Decision variable: Let x_{ij} denote the number of cement bags transported from depot i to destination j. There are $4 \times 5 = 20$ shipment decision variables.

Parameters and Input Data

Let c_{ij} be the transportation cost per cement bag from depot i to destination j, s_i be the supply available at depot i, and d_j be the demand at destination j.

Shipping cost per bag:

Table 2: Shipping Cost per Cement Bag (NGN)

Depot	Ikeja	Ajagunle	Apapa	Idi-Araba	Agege
A	68	36	26	31	62
B	35	72	36	28	54
C	40	55	41	30	28
D	25	35	40	80	34

Supply (cement bags):

Table 3: Available Supply by Depot

Depot	**Supply**
A	1,100
B	1,220
C	950
D	1,830
Total	5,100

Demand (cement bags):

Table 4: Required Demand by Destination

Destination	**Demand**
Ikeja	1,010
Ajagunle	970
Apapa	980
Idi-Araba	1,250
Agege	890
Total	5,100

Total supply equals total demand, so the transportation problem is balanced.

Mathematical Formulation

Objective function

Minimize total transportation cost:

$$\min Z = \sum_{i=1}^{4} \sum_{j=1}^{5} c_{ij} x_{ij}.$$

Supply constraints

All available supply is shipped:

$$\sum_{j} x_{Aj} = 1,100,$$
$$\sum_{j} x_{Bj} = 1,220,$$
$$\sum_{j} x_{Cj} = 950,$$
$$\sum_{j} x_{Dj} = 1,830.$$

Demand constraints

Each destination receives its required demand:

$$\sum_i x_{i,Ikeja} = 1,010,$$
$$\sum_i x_{i,Ajagunle} = 970,$$
$$\sum_i x_{i,Apapa} = 980,$$
$$\sum_i x_{i,Idi-Araba} = 1,250,$$
$$\sum_i x_{i,Agege} = 890.$$

Non-negativity constraints

$$x_{ij} \geq 0.$$

Since total supply equals total demand, the model is balanced.

Initial-Solution Methods

The assigned case compares four methods for generating initial basic feasible solutions: Northwest Corner Method (NWC), Vogel's Approximation Method (VAM), Least Cost Method (LCM), and Row Minima Method (RMM). These methods provide feasible starting allocations, but an optimality procedure is required to determine whether further cost reduction is possible.

Table 5: Reported Initial and Final Transportation Costs

Method	**Initial Cost (NGN)**	**Iterations**	**Final Cost (NGN)**
Northwest Corner	288,520	9	145,140
Vogel Approximation	145,380	2	145,140
Least Cost	160,750	3	145,140
Row Minima	146,760	4	145,140

The reported VAM starting solution is especially close to the reported optimum, with a difference of only NGN 240. This makes VAM the principal starting solution selected for independent replication in this paper.

Optimal Solution

The independently replicated optimal shipping plan is:

Table 6: Replicated Optimal Cement Distribution Plan (bags)

From / To	Ikeja	Ajagunle	Apapa	Idi-Araba	Agege	Total Supply
A	0	120	980	0	0	1,100
B	30	0	0	1,190	0	1,220
C	0	0	0	60	890	950
D	980	850	0	0	0	1,830
Demand	1,010	970	980	1,250	890	5,100

Depot A ships 120 bags to Ajagunle and 980 bags to Apapa. Depot B ships 30 bags to Ikeja and 1,190 bags to Idi-Araba. Depot C ships 60 bags to Idi-Araba and 890 bags to Agege. Depot D ships 980 bags to Ikeja and 850 bags to Ajagunle. All other routes carry zero bags. The plan ships 5,100 bags and satisfies every destination's demand exactly.

Minimum Transportation Cost

Total transportation cost is calculated by multiplying the quantity shipped on each utilized route by its transportation cost per bag and summing the resulting route costs:

$$\begin{aligned} Z &= 120(36) + 980(26) + 30(35) + 1,190(28) \\ &\quad + 60(30) + 890(28) + 980(25) + 850(35) \\ &= 145,140 \text{ NGN}. \end{aligned}$$

Thus, the independently replicated minimum transportation cost is NGN 145,140.

Solution Method

The cement-distribution problem is a linear programming transportation model. The objective is to minimize total shipping cost while meeting each destination's demand and using the available supply at each depot. The source compares NWC, VAM, LCM, and RMM for initial feasible solutions and then evaluates the solutions using an optimality procedure. In this replication, the VAM allocation is checked using the Modified Distribution Method (MODI).

Model Implementation

The model contains 20 decision variables, x_{ij}, where each variable represents the number of cement bags shipped from one of the four depots to one of the five destinations. The objective function uses the same cost matrix reported in the case study. The supply constraints require the row totals to equal 1,100, 1,220, 950, and 1,830 bags, respectively. The demand constraints require the column totals to equal 1,010, 970, 980, 1,250, and 890 bags, respectively. Non-negativity is imposed on all shipment variables.

For computational replication in a spreadsheet, the cost matrix and shipment matrix should be entered separately. The shipment matrix must remain the changing-variable range; the reported

optimal allocation must not be entered as a fixed constraint. The objective cell should calculate total transportation cost with a SUMPRODUCT formula, while row and column totals should be used for the supply and demand constraints.

Spreadsheet Solver Verification

To support the computational replication, the transportation model can be implemented in a spreadsheet Solver using the same cost matrix, supply values, demand values, and non-negativity conditions defined in the mathematical model. The shipment matrix is treated as the changing-variable range, while the total transportation cost is the objective cell to be minimized. The Solver setup is summarized below.

Table 7: Solver Parameters for Spreadsheet Replication

Solver Item	**Required Setting**
Objective Cell	B15
Optimization	Minimize
Changing Variable Cells	B8:F11
Supply Constraints	G8:G11 = H8:H11
Demand Constraints	B12:F12 = B13:F13
Variable Restrictions	Shipment quantities are non-negative
Solving Method	Simplex LP
Objective Formula	`=SUMPRODUCT(B2:F5,B8:F11)`

The Solver solution should return a minimum transportation cost of NGN 145,140. The resulting shipment quantities should satisfy all supply and demand constraints. The expected optimal allocation is shown in Table 6, and the resulting objective value provides an independent spreadsheet-based confirmation of the MODI result.

Table 8: Solver Result for the Transportation Model

Solver Result	**Value/Status**
Final objective value	NGN 145,140
Optimization status	Minimum transportation cost
Supply constraints	Satisfied
Demand constraints	Satisfied
Non-negativity	Satisfied
Solving method	Simplex LP
Final allocation	Same optimal allocation shown in Table 6

For software evidence, the Solver Parameters and Solver Results screens should be captured after solving the model. These screenshots document the spreadsheet implementation and confirm that the reported minimum cost is obtained while all model constraints are satisfied.

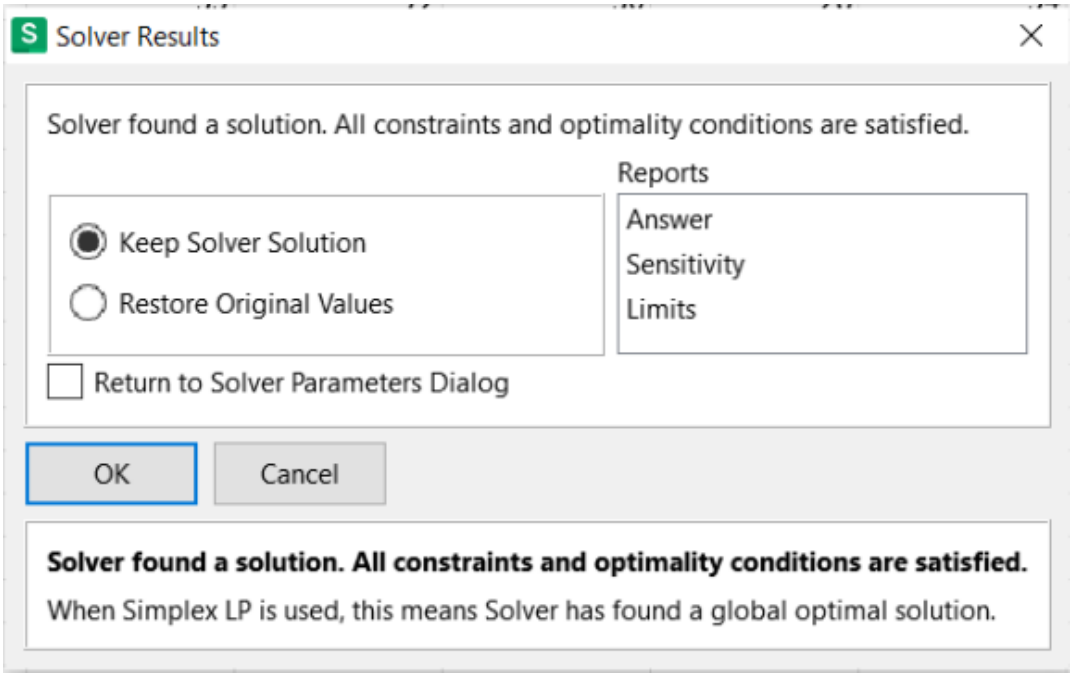


Figure 1: WPS Solver Results showing that all constraints and optimality conditions are satisfied.

Transportation Cost	Ikeja	Ajagunle	Apapa	Idi-Araba	Agege	Row Total	Supply
A	68	36	26	31	62	223	1100
B	35	72	36	28	54	225	1220
C	40	55	41	30	28	194	950
D	25	35	40	80	34	214	1830
Decision Variables	Ikeja	Ajagunle	Apapa	Idi-Araba	Agege	Row Total	Supply
A	0	120	980	0	0	1100	1100
B	30	0	0	1190	0	1220	1220
C	0	0	0	60	890	950	950
D	980	850	0	0	0	1830	1830
Demand Total	1010	970	980	1250	890		
Demand	1010	970	980	1250	890		
Total Transportation Cost	145140						

Figure 2: Final WPS Solver allocation showing the optimal shipment quantities and minimum transportation cost of NGN 145,140.

Replicated VAM Starting Allocation

The VAM starting allocation reported for the case can be independently reproduced as follows:

Table 9: Replicated VAM Starting Allocation (bags)

Depot	Ikeja	Ajagunle	Apapa	Idi-Araba	Agege	Total Supply
A	0	120	980	0	0	1,100
B	0	0	0	1,220	0	1,220
C	0	30	0	30	890	950
D	1,010	820	0	0	0	1,830
Demand	1,010	970	980	1,250	890	5,100

Its total cost is:

$$\begin{aligned} Z &= 120(36) + 980(26) + 1,220(28) + 30(55) + 30(30) \\ &\quad + 890(28) + 1,010(25) + 820(35) \\ &= 145,380 \text{ NGN}. \end{aligned}$$

MODI Optimality Check

The Modified Distribution Method (MODI) was used to test the replicated VAM allocation. For occupied cells, the potentials satisfy

$$u_i + v_j = c_{ij}.$$

For an unoccupied cell, the reduced cost is

$$\Delta_{ij} = c_{ij} - (u_i + v_j).$$

The VAM solution contains an improving route with $\Delta_{BI} = -8$, indicating that the starting solution is not yet optimal. The improvement loop has $\theta = 30$ bags. Applying the loop produces the optimal allocation shown previously, with total transportation cost of NGN 145,140. Therefore, the improvement from the VAM starting solution is

$$145,380 - 145,140 = 240 \text{ NGN}.$$

Feasibility Verification

The allocation is feasible because every depot total equals its available supply and every destination receives its required demand.

Supply totals:

- Depot A: $120 + 980 = 1,100$ bags
- Depot B: $30 + 1,190 = 1,220$ bags
- Depot C: $60 + 890 = 950$ bags
- Depot D: $980 + 850 = 1,830$ bags

Destination demand totals:

- Ikeja: $30 + 980 = 1,010$ bags

- Ajagunle: $120 + 850 = 970$ bags
- Apapa: 980 bags
- Idi-Araba: $1,190 + 60 = 1,250$ bags
- Agege: 890 bags

All shipment quantities are non-negative. The total shipped equals both total supply and total demand, so the allocation satisfies all model constraints.

Comparison with the Reported Case Study

The reported case-study result and the independent replication are compared below.

Table 10: Comparison of Reported and Replicated Results

Evaluation Criteria	**Case Study Reported Result**	**Independent Replication**
Total supply	5,100 bags	5,100 bags
Total demand	5,100 bags	5,100 bags
VAM initial cost	NGN 145,380	NGN 145,380
Final transportation cost	NGN 145,140	NGN 145,140
Supply constraints	Satisfied	Satisfied
Demand constraints	Satisfied	Satisfied
Optimality	Reported as optimal	Confirmed through MODI
Improvement from VAM	NGN 240	NGN 240

The allocation and minimum transportation cost agree with the reported case-study result. The independent replication therefore supports the reported optimum of NGN 145,140. The VAM starting solution also agrees exactly with the reported NGN 145,380 value.

Discussion of the Replication Results

The model identifies a cement distribution plan that satisfies all five destination demands while using all 5,100 available bags and minimizing transportation cost under the stated route-cost matrix. The replicated optimum of NGN 145,140 confirms the central numerical result reported in the assigned case. VAM provides an efficient starting point for this particular dataset because its initial cost is only NGN 240 above the independently verified optimum.

The result depends on the stated input data and assumptions. In particular, the model treats transportation costs as fixed, assumes that all listed supply is available, and requires each destination's demand to be met exactly. The classical transportation model also does not directly account for delivery time, vehicle capacity, fuel consumption, congestion, road conditions, service level, or environmental

considerations. Consequently, the model is best interpreted as a cost-minimization baseline for the distribution decision.

A numerical inconsistency in the source material should also be acknowledged. The assigned case reports NGN 160,750 for the Least Cost Method in its comparison table, while another statement associated with the underlying MAISANGO report gives NGN 146,750. This paper retains NGN 160,750 for the comparison because it is the value presented in the assigned case's comparative table. The discrepancy does not affect the replicated VAM cost or the final optimum of NGN 145,140.

Conclusion

In conclusion, the Transportation Model under Linear Programming provides a systematic method for determining an economical distribution plan for MAISANGO Cement Nigeria Plc. The case consists of four depots and five destinations, with total supply and total demand both equal to 5,100 cement bags. The objective is to minimize transportation cost while satisfying every supply and demand constraint.

The independent replication confirms the VAM starting cost of NGN 145,380 and the final minimum transportation cost of NGN 145,140 after a MODI improvement of NGN 240. The replicated allocation is feasible and agrees with the reported optimum in the case study. The analysis demonstrates the value of Operations Research in converting a distribution problem into a transparent quantitative model, while also showing why source data, assumptions, and optimality claims must be checked before applying a model to actual logistics operations.

References

Abdullahi, I., Kankarofi, R. H., Yakubu, U. A., Mandara, A. V., & Murtala, S. (2020). Minimizing cost of transportation by Vogel's approximation method: A case study of MAISANGO Cement Nigeria Plc. *Malaysian Journal of Computing and Applied Mathematics, 3*(2), 36–44. https://doi.org/10.37231/myjcam.2020.3.2.50

Mehta, A., Kasera, A., Sane, A., Misra, A., & Gadiyar, A. (2026). Transportation models for cost minimization in distribution logistics. *International Journal of Science, Strategic Management and Technology, 2*(9), 1–9. https://doi.org/10.55041/ijsmt.v2i9.022

Mula, J., Peidro, D., Díaz-Madroñero, M., & Vicens, E. (2010). Mathematical programming models for supply chain production and transport planning. *European Journal of Operational Research, 204*(3), 377–390. https://doi.org/10.1016/j.ejor.2009.09.008

Nursafwah, Matondang, N., & Ishak, A. (2023). A literature review of distribution solving problems by various transportation methods. *Jurnal Sistem Teknik Industri, 25*(2), 178–187. https://doi.org/10.32734/jsti.v25i2.9741

Rafi, F. S., & Islam, S. (2020). A comparative study of solving methods of transportation problem in linear programming problem. *Journal of Advances in Mathematics and Computer Science, 35*(5), 45–67. https://doi.org/10.9734/jamcs/2020/v35i530281

Shore, H. H. (1970). The transportation problem and the Vogel approximation method. *Decision Sciences, 1*(3–4), 441–457. https://doi.org/10.1111/j.1540-5915.1970.tb00792.x

Zhang, N. L., & Xie, F. R. (2025). A review of research on the transportation problem. *Open Journal of Applied Sciences, 15*, 1168–1177. https://doi.org/10.4236/ojapps.2025.155081